

Task ①: {using inverting op-amp logic}

$$\begin{aligned}
 \bullet) V_A &= -V_1 \left(\frac{R_2}{R_1} \right) & \bullet) V_B &= -V_2 \left(\frac{R_4}{R_3} \right) & \bullet) V_C &= - \left[\frac{R_8 V_A}{R_5} + \frac{R_3 V_B}{R_6} + \frac{R_8 V_3}{R_7} \right] \\
 \bullet) V_D &= - \left[\frac{R_{11} V_4}{R_9} + \frac{R_{11} V_5}{R_{10}} \right] & V_C &= \left[\frac{R_2 R_8 V_1}{R_5 R_1} + \frac{R_4 R_8 V_2}{R_3 R_6} - \frac{R_8 V_3}{R_7} \right] \\
 \bullet) V_E &= - \left[\frac{R_{14} V_D}{R_{12}} + \frac{R_{14} V_C}{R_{13}} \right] = \frac{R_{11} R_{14} V_4}{R_9 R_{12}} + \frac{R_{11} R_{14} V_5}{R_{10} R_{12}} - \frac{R_{14} V_C}{R_{13}} \\
 \bullet) V_F &= - \left[\frac{R_{17} V_C}{R_{15}} + \frac{R_{17} V_E}{R_{16}} \right] = V_{out} \\
 V_F &= - \left[\frac{R_{17}}{R_{15}} \left[\frac{R_2 R_8 V_1}{R_5 R_1} + \frac{R_4 R_8 V_2}{R_3 R_6} - \frac{R_8 V_3}{R_7} \right] + \frac{R_{17}}{R_{16}} \left[\frac{R_{11} R_{14} V_4}{R_9 R_{12}} + \frac{R_{11} R_{14} V_5}{R_{10} R_{12}} - \frac{R_{14} V_C}{R_{13}} \right] \right] \\
 V_F &= \frac{-2 R_{17} R_2 R_8}{R_{15} R_5 R_1} - \frac{R_{17} R_4 R_8}{R_{15} R_3 R_6} + \frac{2 R_{17} R_8}{R_{15} R_7} + \frac{2 R_{17} R_{14} R_{11}}{R_{16} R_{12} R_9} + \frac{R_{17} R_{14} R_{11}}{2 R_{12} R_{10}} + \frac{5 R_{17} R_{14}}{2 R_{16} R_{13}}
 \end{aligned}$$

Task (2): [ROUGH WORKING]

1) $V_i = 2V$, $V_A = -2V$, $R_1 = R_2$ because $V_A = -V_i \left(\frac{R_2}{R_1} \right)$

$$V_f = -\frac{2R_1R_8}{R_{15}R_5} - \frac{R_1R_4R_8}{R_{15}R_3R_6} + \frac{2R_1R_8}{R_{15}R_7} + \frac{R_1R_{14}R_{11}}{R_{15}R_9R_9} + \frac{R_1R_{14}R_{11}}{(20k)R_{10}} + \frac{5R_1R_{14}}{2R_1R_8}$$

2) we can now assume values of R_1 and R_2 , as long as they are equal. In our case, $R_1 = R_2 = 1k\Omega$.

3) $V_f = -R_{17} \left(\frac{V_c}{R_{15}} + \frac{V_E}{R_{16}} \right)$

$12 = -R_{17} \left(\frac{V_c}{R_{15}} + \frac{V_E}{R_{16}} \right) \Rightarrow$ if we assume $R_{17} = 12k\Omega$, then $\left(\frac{V_c}{R_{15}} + \frac{V_E}{R_{16}} \right) = -1$

4) now, $\frac{V_c}{R_{15}} + \frac{V_E}{R_{16}} = -1$

we can assume $\frac{V_c}{R_{15}} = -2$ and $\frac{V_E}{R_{16}} = +1$

$$\frac{2R_8}{R_{15}R_5} - \frac{R_4R_8}{R_{15}R_3R_6} - \frac{2R_8}{R_{15}R_7} = -2$$

$$\frac{R_8}{R_{15}} \left[\frac{2}{R_5} - \frac{R_4}{R_3R_6} - \frac{2}{R_7} \right] = -2$$

we can assume this ratio is 2 $\Rightarrow R_8 = 2k\Omega$, $R_{15} = 1k\Omega$

we can assume this all sums up to -1

$$\frac{R_{14}}{R_{16}} \left[\frac{2R_{11}}{R_9(10k)} + \frac{R_{11}}{(20k)R_{10}} + \frac{5}{2R_{13}} \right] = 1$$

we can assume this ratio of $\frac{R_{14}}{R_{16}} = 1$

$R_{14} = R_{16} = 10k\Omega$

we can assume this all adds up to 1.

$$\frac{2}{R_5} - \frac{R_4}{R_3 R_6} - \frac{2}{R_7} = -1$$

if $R_5 = R_7$, then,

$$\frac{2}{R_5} - \frac{2}{R_7} - \frac{R_4}{R_3 R_6} = -1$$

$$\frac{R_4}{R_3 R_6} = +1$$

now we just need for $\frac{R_4}{R_3 R_6}$ to be 1, so we can assume

$$R_4 = R_3 = R_6 = 1k\Omega$$

$$\frac{R_8}{R_{15}} \left[\frac{2}{R_5} - \frac{R_4}{R_3 R_6} - \frac{2}{R_7} \right]$$

$$+ \frac{R_{11}}{R_{16}} \left[\frac{2 R_{11}}{(10k)(R_9)} + \frac{R_{11} + 5}{(20k)R_{10} \frac{1}{2R_{13}}} \right] 4x + y = 5xy$$

$$= \frac{2}{1} \left[\frac{2}{1} - \frac{1}{(1)(1)} \right] = \frac{-2}{10}$$

$$\frac{R_{11}}{(5k)R_9} + \frac{R_{11}}{(20k)R_{10}} + \frac{5}{2R_{13}} = 1$$

$$R_{11} \left[\frac{1}{(5k)R_9} + \frac{1}{(20k)R_{10}} \right] + \frac{5}{2R_{13}} = 1$$

$$R_{11} \left[\frac{1}{5kR_9} + \frac{1}{20kR_{10}} \right] = 0.25 \quad \frac{5}{2R_{13}} = 0.75$$

$$1k\Omega \quad 0.25$$

$$5 = 15k\Omega$$

$$R_{13} = 3.3k\Omega$$

$$\frac{1}{5000 R_9} + \frac{1}{20000 R_{10}} = 0.25$$

$$20k R_{10} + 5k R_9 = 0.25$$

$$100k R_9 R_{10}$$

$$20k R_{10} + 5k R_9 = 25k R_9 R_{10}$$

$$4k R_{10} + R_9 = 5k R_9 R_{10}$$

$$R_{10} = R_9 = 1$$

$$\begin{array}{ll} R_1 = 1 & R_9 = 1 \\ R_2 = 1 & R_{10} = 1 \\ R_3 = 1 & R_{11} = 1 \\ R_4 = 1 & R_{12} = 10 \end{array}$$

Task (2)

$$\Rightarrow V_1 = 2V \Rightarrow V_A = -2V \Rightarrow R_1 = R_2 \text{ because } V_A = -V_1 \left(\frac{R_2}{R_1} \right)$$

After trial and error, + some simulation on PSPICE,
if we take $R_1 = R_2 = R_3 = R_5 = R_6 = R_8 = R_9 = R_{10} = R_{11} = R_{12} = R_{13} = R_{14} = R_{15} = R_{16} = 1k$,
 $R_4 = R_7 = 2k$, $R_{17} = 6\Omega$, we can verify that:

$$V_F = 12 = -2 \left(\frac{6 \cdot 1 \cdot 1}{1 \cdot 1 \cdot 1} \right) - \left(\frac{6 \cdot 2 \cdot 1}{1 \cdot 1 \cdot 1} \right) + 2 \left(\frac{6 \cdot 1}{1 \cdot 2} \right) + 2 \left(\frac{6 \cdot 1 \cdot 1}{1 \cdot 1 \cdot 1} \right) + \left(\frac{6 \cdot 1 \cdot 1}{2 \cdot 1 \cdot 1} \right) + 5 \left(\frac{6 \cdot 1}{2 \cdot 1 \cdot 1} \right)$$

$$12 = -2 - 12 + 6 + 12 + 3 + 15$$

$$12 = 12$$

Final values:

$$\Rightarrow R_1 = R_2 = R_3 = R_5 = R_6 = R_8 = R_9 = R_{10} = R_{11} = R_{12} = R_{13} = R_{14} = R_{15} = R_{16} = 1k$$

$$\Rightarrow R_4 = R_7 = 2k \Rightarrow R_{17} = 6$$

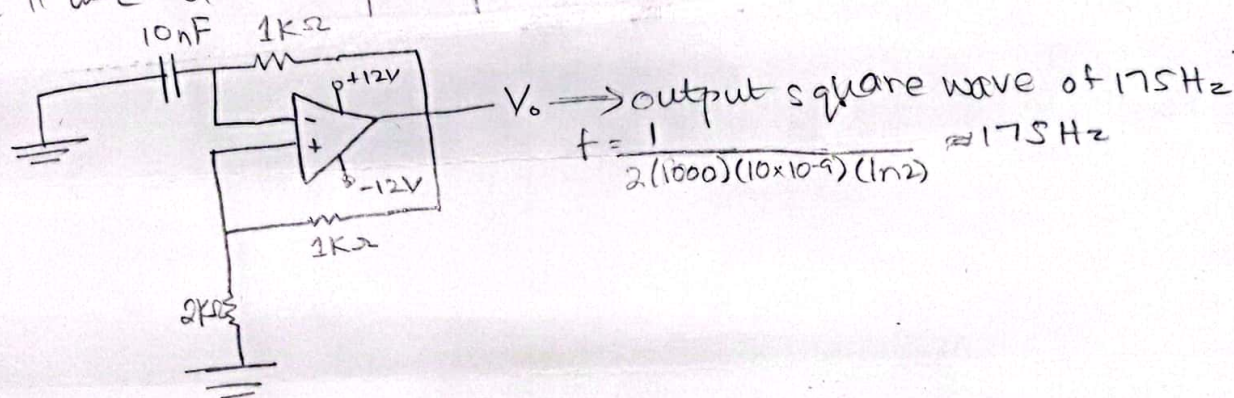
$$\Rightarrow V_A = -2V \Rightarrow V_B = -2V \Rightarrow V_C = 3V \Rightarrow V_D = 2.5V \Rightarrow V_E = 5V$$

$$\Rightarrow V_F = 12V$$

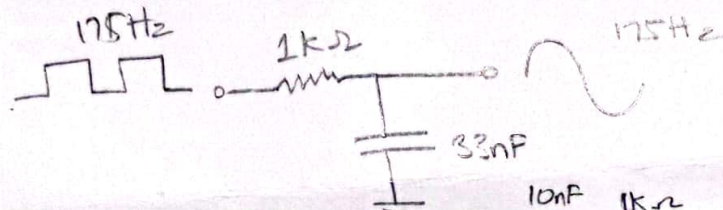
Task (3):

step (1): square wave generation using an op-amp.

we'll use a 741-op amp.



step (2): square wave to sine wave converter.



Final Block (2) diagram:

